

Split Complex Lorentzian Dynamics

Part XXXII: Ordinary-Atlas Descent, the Bare-Family Detour, and the First Cross-Experiment Interface

Why Task XXXII closes the presentation layer but removes essential topology, and why Experiment 1 now supplies the missing geometric carrier

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Abstract

Part XXXI ended with a precise warning. Task 31 had solved the internal lifting problem for a *fixed ordinary transition presentation*: compatible continuous internal transitions exist if and only if the reconstructed kernel-defect state is neutral. It also supplied a non-neutral presentation. Source review then showed that the negative example lived on a constant, globally trivializable metric-oriented rank-three family and became non-liftable only after a redundant second global ordinary chart was frozen. The next task therefore had to distinguish presentation-level non-liftability from geometric non-liftability and quotient ordinary chart changes and ordinary cover refinements before any spatial spin-structure claim could be made [15].

Task 32 performs a substantial part of exactly that programme. It separates a bare metric-oriented family from an ordinary atlas presentation, derives the ordinary chart-gauge law

$$g'_{ij} = h_j g_{ij} h_i^{-1},$$

constructs ordinary cover refinements and common refinements, packages a generated atlas-equivalence relation, and formalizes the mandatory Task-31 good/bad control. The same underlying constant family is proved to admit both a directly liftable neutral presentation and a non-liftable non-neutral presentation. The chart gauge connecting them is explicitly $h_0 = 1$, $h_1(b) = b$ and is proved not to admit the required global internal lift. Direct liftability passes to refinements, while the converse is explicitly refuted. These are genuine theorem-level advances.

The task then defines

$$\text{FamilyLiftable}(F) :\iff \exists \text{ an ordinary atlas } A \text{ of } F \text{ with } \text{DirectAtlasLiftable}(A)$$

and an analogous existential family-level defect-neutrality predicate. It proves the exact formal equivalence

$$\boxed{\text{FamilyLiftable}(F) \iff \text{FamilyKernelDefectNeutral}(F)}$$

and records formal verdict C: exact family-level classification, but no universal positive theorem and no genuine family-level negative example [1].

The source-level review in the present paper finds that this descent goes one level too far. The object called `BareMetricOrientedFamily` is definitionally only the Task-XXVI `MetricOrientedRankThreeFamily`: it stores a three-dimensional inner-product space and an orientation in each fibre, but no topology on the total carrier. Task XXVI had explicitly proved that fibrewise algebraic trivializations exist over every subset by Choice while providing no regularity in the base point, and had explicitly identified the correct bundle topology as missing input. Task XXVII then introduced an atlas precisely to generate that topology and proved uniqueness only relative to that atlas. Therefore the passage from `TopologicalFrameFamily` to the bare family removes not only arbitrary chart presentation but also the topological gluing structure that had made the family a topological geometric object.

This explains why the Task-32 family-level invariance theorems are formally true yet too weak for the intended geometric question. The theorem named `familyKernelDefectNeutral_atlasEquivalence_invariant`, for example, does not consume its atlas-equivalence proof: existential family neutrality follows merely because either liftable atlas is an existential witness. More strongly, the inherited Task-XXVI Choice theorem strongly indicates a short one-chart construction proving *universal FamilyLiftable* under the current definition: choose one algebraic trivialization on the whole base, use it as the sole chart, obtain the identity transition, and invoke the already certified trivial-transition lift. This is a source-level derivation, not a newly compiled Lean theorem in the supplied artifact, and is therefore recorded as a hardening target rather than silently promoted. If closed, it would give formal verdict A at the current abstraction level for a reason that is geometrically uninformative: the topology has been allowed to be regenerated from the chosen global algebraic chart.

At this point the correct next question is no longer purely internal to Experiment 2. Experiment 1 independently decomposed the same split-complex Lorentz structure and already reconstructed a positive Euclidean three-space, a Lorentz spin factor, the future cone, the proper Lorentz group, the future unit shell $\mathcal{H}$, the shell tangent metric, its intrinsic distance and topology, Hodge structure, Clifford structure, and an intrinsic spin-cover branch. In particular it proved

$$T_u \mathcal{H} = u^{\perp_{B_S}}, \quad g_u(v, w) = -B_S(v, w),$$

with g_u positive definite, and recovered the shell topology independently from the intrinsic distance [12, 13, 14]. Established mathematics identifies this shell with hyperbolic three-space and gives the exact homogeneous-space and frame-bundle relations

$$\mathcal{H} \cong H^3 \cong SO^+(1, 3)/SO(3) \cong SL(2, \mathbb{C})/SU(2),$$

$$\text{Fr}^+(H^3) \cong SO^+(1, 3), \quad \text{Spin}^+(1, 3) \longrightarrow SO^+(1, 3)$$

with stabilizers $SO(3)$ and $\text{Spin}(3) \cong SU(2)$ and kernel $\{\pm 1\}$ [9, 10, 5, 2]. These are exactly the established objects independently approached by Experiment 2: a metric-oriented three-dimensional spatial fibre, a visible $SO(3)$ -type action, an $SU(2)$ -type internal lift, and a two-element central kernel.

The consequence is important but must be stated at the correct proof level. Experiment 2 now appears to require topological and geometric structure that Experiment 1 already produces naturally from the common Lorentz starting object. This is the first point at which a controlled cross-experiment interface is not merely interesting but may be necessary to avoid continuing with an overgeneralized arbitrary-family theory. It is *not* yet a formal merger. The public Experiment-1 branch currently exposes papers and a compressed Aristotle archive rather than an unpacked Lake package, while the Experiment-2 Task-32 artifact is an unpacked Lake project. Lake itself supports local-path and Git dependencies, so cross-package imports are standard at the Lean build-system level [7, 6]. Whether the Aristotle project environment can consume another project in the required form has not yet been tested. Refactoring may therefore be required before mathematics can continue: frozen E1 and E2 sources should be exported behind narrow, differently named interface packages, and a tiny bridge project should first test one dependency in each direction without importing the full historical payload.

Part XXXII therefore records a productive detour rather than a failure. Aristotle correctly solved the problem it was given at the bare-family level; source review shows that this level had discarded essential topology. That discovery in turn identifies the first mathematically compelled connection to Experiment 1. The next step is no longer “prove one more obstruction theorem.” It is an engineering and dependency-boundary test: determine whether the two formally independent excavations can be imported together without contaminating their proof histories, expose only the minimal common interfaces, and then prove whether the E1 shell/rest-space geometry instantiates the E2 spatial frame/lift architecture. The excavation has reached a junction rather than bedrock.

Keywords: atlas descent; total-space topology; metric-oriented frame family; hyperbolic shell; Lorentz stabilizer; $SO(3)$; $SU(2)$; spin lift; cross-experiment interface; Lake package dependency; Lean 4; Aristotle; Conservation of Difficulty.

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1 The question inherited from Part XXXI

Part XXXI had separated two statements that had initially looked identical:

Direct fixed-presentation liftability.

Lift every chosen g_{ij} continuously on the chosen whole overlaps.

Intrinsic geometric liftability.

The underlying geometry admits a global upstairs object after admissible presentation changes.

The Task-31 bad system proved that the first can fail. The source review then observed that its underlying rank-three family was constant and globally trivializable. Therefore the second statement had not been refuted.

Task 32 was consequently designed to remove three kinds of auxiliary freedom:

internal kernel gauges + ordinary chart gauges + ordinary cover refinements.

The expected endpoint was an atlas-independent family-level defect or at least an atlas-independent family-level neutrality predicate. This was the correct next question. The subtlety exposed by the result is that an atlas can carry both *presentation redundancy* and *essential topological gluing information*. Quotienting the former must not delete the latter.

Methodological principle 1 (Presentation may contain geometry). An object should not be called “bare geometry” merely because all local coordinates have been removed. If the removed data were also the only carrier of the total-space topology, then the quotient has erased geometric structure rather than only presentation structure.

2 Formal Task-XXXII source, build status, and provenance

The supplied Task-32 Aristotle archive has SHA-256

c0098d23d0741123043e5aedd570d32566855e5f5ac29ad7b6a0a0f9f340321e.

It uses Lean 4.28.0 and requires Mathlib v4.28.0, resolved in the manifest to commit

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The supplied audit reports a focused green Task-32 build and a whole-project green build of 8360 jobs, with no new Task-32 warnings and only the baseline axioms `propext`, `Classical.choice`, and `Quot.sound`. Direct inspection finds no `sorry`, `admit`, custom axiom, `implemented_by`, `native_decide`, `bv_decide`, prohibited `unsafe`, or prohibited `partial` in the new layer.

The new source directory contains ten Lean files and 1871 Lean source lines:

UnderlyingFrameGeometry → OrdinaryAtlasChange → OrdinaryCoverRefinement →
OrdinaryAtlasEquivalence → FamilyLiftability → AtlasChangeLiftability →
GoodBadAtlasComparison → FamilyDefectNeutrality → FamilyExistenceClassification → Task32.

The Task-32 audit and README say “9 modules”; the actual archive contains ten Lean files including the endpoint file `Task32.lean`. This is a documentation count error, not a mathematical defect.

For provenance, the common Lean sources from the preceding Task-31 archive were compared against Task 32: 324 pre-existing Lean files are byte-identical, with zero changed and zero missing files. Task 32 is therefore an additive layer over the frozen earlier proof tree.

Proof boundary 1 (Verification boundary). *The Task-32 source tree, principal definitions, good/bad control, refinement logic, family-level predicates, audit and build metadata were inspected directly. The reported 8360-job build was not independently rerun in the present paper-generation environment because Lean/Lake executables are not installed here. Source-level deductions introduced below are labelled explicitly and are not presented as compiled Task-32 theorems.*

3 What Task XXXII closes correctly: ordinary chart change

The first major result is exactly the one Part XXXI requested. Task 32 separates a bare family from an ordinary atlas presentation and defines the chart-comparison map between two ordinary metric-oriented trivializations. The transition convention is not guessed; the transformation law is derived:

$$g'_{ij}(b) = h_j(b) g_{ij}(b) h_i(b)^{-1}. \quad (1)$$

Identity, inverse and composition of chart changes are proved, as is continuity of the comparison maps for a joint ordinary atlas. The usual cocycle compatibility is therefore recovered internally rather than imported from bundle theory.

This is the first place where Task 32 succeeds in exactly the intended sense: ordinary chart choice is made explicit as a transformation of presentation data while the underlying pointwise metric-oriented fibres remain unchanged.

4 Ordinary cover refinement and the generated atlas relation

Task 32 also distinguishes ordinary cover refinement from the Task-XXVIII/XXIX refinement of local internal representatives. An `OrdinaryCoverRefinement` carries a new index type, a map to the old chart index, open refined domains, a cover proof and subset inclusions. The fibre trivializations are restricted, and the refined transition functions are proved to be literal restrictions of the old transition functions.

Identity refinement and composition are proved, and intersection covers provide common refinements of two ordinary atlases. A generated finite-zig-zag relation `OrdinaryAtlasEquivalent` is then constructed from chart-gauge steps, refinement steps, symmetry and transitivity.

At this level the task has reconstructed a recognizable fragment of standard atlas equivalence without importing a vector-bundle category. This is mathematically useful even though, as shown later, the relation has not yet been tied to a fixed total-space topology.

5 The mandatory good/bad control is now a Lean theorem

The most important formal success of Task 32 is the repair of the Task-31 counterexample. The two atlases are now theoremtically placed over the same bare constant family. The ordinary gauge between them satisfies

$$h_0(b) = 1, \quad h_1(b) = b.$$

The trivial presentation is directly liftable and has neutral Task-31 defect. The two-chart presentation is not directly liftable and has non-neutral Task-31 defect. The two atlases are ordinary-chart-gauge equivalent and also related by the generated atlas-equivalence relation.

Hence Task 32 proves the precise negative control

Same bare family + different ordinary presentations $\implies$ different
fixed-presentation liftability can occur.

This theorem is stronger and cleaner than the source-level observation of Part XXXI because the relation between the two presentations is now part of the formal layer.

Task 32 goes further. It defines a condition saying that an ordinary chart gauge itself admits compatible internal lifts. Under this stronger hypothesis direct liftability transfers. The good/bad gauge fails that hypothesis because $h_1(b) = b$ would require a continuous global representative of the identity map of the visible group. Thus the formalism now explains *why* direct liftability changes under the ordinary chart transformation.

6 Refinement exposes the same distinction a second time

Direct liftability passes from an atlas to every ordinary cover refinement by restricting already compatible internal transitions. The converse is explicitly refuted. The Task-31 bad atlas has a refinement selecting only its first global chart; that refinement has trivial transitions and is directly liftable, even though the original two-chart atlas is not.

The resulting hierarchy is now formally unavoidable:

$$\text{DirectAtlasLiftable}(A) \implies \text{RefinementLiftable}(A),$$

but not conversely.

This is an important theorem independently of the later review. It says that whole-overlap liftability is a stronger condition than existence after shrinking/restricting the ordinary cover. In conventional bundle language this is exactly the kind of distinction one expects local trivialization theory to force, but Task 32 obtains it from the reconstructed interfaces themselves.

7 The formal Task-32 family-level classification

Task 32 next defines

$$\text{FamilyLiftable}(F) :\iff \exists \iota \exists A : \text{OrdinaryAtlasPresentation}(F, \iota), \quad \text{DirectAtlasLiftable}(A). \quad (2)$$

It also defines

$$\text{FamilyKernelDefectNeutral}(F) :\iff \exists \iota \exists A, \quad \text{IsNeutralGlobalKernelDefect}(\text{ordinarySystem}(A)). \quad (3)$$

Because Task 31 had already proved direct liftability if and only if fixed-system defect neutrality, Task 32 obtains immediately

$$\boxed{\text{FamilyLiftable}(F) \iff \text{FamilyKernelDefectNeutral}(F).} \quad (4)$$

The Task-31 constant family is correctly classified as family-level positive because its trivial presentation witnesses (2).

The source audit freezes formal verdict C: the equivalence (4) is proved, but universal family liftability is left undecided and no genuinely family-level negative example is produced.

Status 1 (Formal Task-32 verdict). At the exact definitions present in the supplied Lean artifact, verdict C is the declared endpoint: classification by existential family neutrality is proved; universal family liftability is neither proved nor refuted.

8 Review turning point: the “bare family” is not yet topological geometry

The key definition is short:

$$\text{BareMetricOrientedFamilyBE} := \text{MetricOrientedRankThreeFamilyBE}.$$

Task XXVI’s `MetricOrientedRankThreeFamily` contains precisely two additional fields beyond the ambient fibrewise inner-product structures:

1. every fibre has real dimension three;
2. every fibre carries an orientation datum.

Its own source documentation explicitly says: no vector bundle, no principal bundle, no local triviality, and *no topology on any total space*.

This matters because the topology had not been redundant data in the earlier dependency graph. Task XXVI proved that an algebraic orientation-preserving fibre trivialization exists over *every subset* of the base using Choice:

$$\forall U \subseteq B, \quad \text{Nonempty}(\text{LocalFibreTriv}(F, U)).$$

The same source emphasizes that no regularity in b is present or even stateable without a topology on the total fibre carrier. The point of Task XXVII was precisely to add the missing topological interface.

Review finding 1 (The Task-32 quotient removes more than presentation). Passing from `TopologicalFrameFamily` to `BareMetricOrientedFamily` removes the chosen atlas, but it simultaneously removes the only datum from which Task XXVII generated the topology on the total fibre carrier. Therefore `FamilyLiftable` is atlas-free only because it quantifies over all possible atlases that may generate different total-space topologies.

9 Task XXVII had already identified the missing variable

Task XXVII defines a metric-oriented fibre atlas as an open cover, algebraic local fibre trivializations, and continuous overlap maps. From this atlas it constructs a topology

$$\tau_A = \bigvee_i \text{coinduced}(\psi_i, \tau_{U_i \times V})$$

after the appropriate local-product convention, and proves two complementary results:

1. the atlas-generated topology makes the base projection continuous and every chart a homeomorphism;
2. any topology with those properties is equal to the atlas-generated topology.

The uniqueness theorem is therefore *relative to the chosen atlas*. Task XXVII does not prove that arbitrary atlases on the same pointwise family generate the same total-space topology. Indeed, without a prior topological vector-bundle structure that statement need not even be true.

This is exactly the distinction standard bundle theory makes. A topological vector bundle is not merely a set-indexed collection of vector spaces. It includes a topological total space and a continuous projection that is locally a product; compatible bundle charts describe that topology, but do not freely replace it by an arbitrary new one [16, 3].

Methodological principle 2 (Correct descent target). The next atlas-independent object should retain the total-space topology while forgetting the distinguished atlas. Equivalently, it may be represented by an equivalence class of atlases proved to generate the same topology. The pointwise family alone is too weak.

10 Why the Task-32 “invariance” theorem is formally true but geometrically weak

The theorem `familyKernelDefectNeutral_atlasEquivalence_invariant` accepts a proof that two atlases are `OrdinaryAtlasEquivalent`. Its body does not use that proof. The reason is structural rather than accidental: the conclusion is only that a directly liftable atlas on either side provides *some* existential witness to `FamilyKernelDefectNeutralF`. The existential predicate mentions no distinguished atlas, so any witness suffices.

Thus the theorem proves

$$A \text{ liftable} \implies \exists A' \text{ of } F \text{ with neutral defect,}$$

not

$$A \sim A' \implies (\text{defect}(A) \text{ neutral} \iff \text{defect}(A') \text{ neutral}).$$

The latter is in fact false for the Task-31 good/bad pair at the fixed-presentation level.

The important missing theorem is instead something like

$$A \sim A' \implies \tau_A = \tau_{A'}, \quad (5)$$

followed by a definition of liftability relative to the fixed topology represented by that equivalence class.

11 A stronger source-level consequence: current FamilyLiftable is probably universal

Task 32 explicitly leaves

$$\text{UniversalFamilyLiftability} := \forall F, \text{FamilyLiftable}(F)$$

open. The inherited source suggests that, under the current definition, this proposition should be provable by a short construction.

Let F be any `BareMetricOrientedFamily`. Task XXVI gives by Choice a `LocalFibreTriv` over every subset, hence in particular over $U = B$. Take a one-element chart index $\iota = \{*\}$ and define

$$U_* = B, \quad \Phi_* : E_b \simeq V_{\text{model}}$$

using such a global algebraic trivialization. The only overlap is the chart with itself, whose transition is the identity by the inherited `transition\_self` theorem. The identity overlap map is continuous. Thus the one-chart data should form a legitimate Task-XXVII `MetricOrientedFibreAtlas`. Its ordinary transition system has

$$g_{**} = 1,$$

so the inherited trivial-transition theorem gives a compatible internal lift.

This yields the paper-level proof route

$$\begin{aligned} \text{Task-XXVI Choice trivialization on } B &\implies \text{one-chart Task-XXVII atlas} \\ &\implies g_{**} = 1 \\ &\implies \text{DirectAtlasLiftable} \\ &\implies \text{FamilyLiftable}(F). \end{aligned} \quad (6)$$

Proof boundary 2 (Source-level, not compiled in Task 32). *Equation (6) is a source-level deduction from already present definitions and theorems. It was not compiled as a new theorem in the supplied Task-32 project, so this paper does not label `UniversalFamilyLiftability` formally PROVED. It should be the first hardening theorem in the next formal run. If Lean closes the construction without a hidden type/topology mismatch, Task-32 verdict C must be revised to verdict A at the current bare-family abstraction level.*

The scientific interpretation would still be negative for the abstraction, not triumphant for the geometry. The global algebraic chart has been chosen first and the total-space topology is then generated so that this chart is a homeomorphism. A topological obstruction cannot survive if the topology itself is allowed to change with the witness.

12 The productive detour

Task 32 is therefore not a failed task. It proves the ordinary-presentation algebra needed for the next level and simultaneously demonstrates that the attempted family quotient was too aggressive. This is a textbook instance of the project's Conservation-of-Difficulty rule.

The unresolved datum has migrated as follows:

$$\begin{aligned}
 &\text{local representatives} \rightarrow \delta_{ijk} \rightarrow \text{kernel equations} \\
 &\quad \rightarrow \text{compatible transitions} \rightarrow \text{global glued object} \\
 &\quad \rightarrow \text{fixed-atlas neutrality} \rightarrow \text{ordinary atlas descent} \\
 &\quad \rightarrow \boxed{\text{fixed total-space topology / topological bundle structure}}.
 \end{aligned}$$

The last box is not an arbitrary technicality. It is exactly the information Task XXVI had declared absent and Task XXVII had introduced.

At this point continuing to develop arbitrary bare rank-three families inside E2 risks drifting into general vector-bundle/spin-bundle theory while losing contact with the original common Lorentz starting object. This motivates the first controlled inspection of Experiment 1.

13 Why Experiment 1 becomes relevant exactly now

Experiment 1 and Experiment 2 began from the same selected split-complex Lorentz structure but were intentionally kept independent. Experiment 2 did not know the Experiment-1 decomposition path. This firewall was essential: if E2 now reconstructs objects already present in E1, the agreement is evidentially meaningful because it was not supplied as input.

Experiment 1's final dependency graph does not stop at the Lorentz form. It descends to a positive Euclidean three-space

$$V = \mathbb{R}^3, \quad \langle \cdot, \cdot \rangle_V > 0,$$

constructs the real spin factor

$$\mathcal{S} = \mathbb{R} \oplus V,$$

then derives the Lorentz norm $N_{\mathcal{S}}$, polarization $B_{\mathcal{S}}$, future cone and intrinsic proper Lorentz group G_L . Downstream branches reconstruct causal order, Alexandrov topology, the future unit shell $\mathcal{H}$, shell metric geometry, Hodge structure, Clifford structure and a spin cover [13, 14].

The relevance to the E2 impasse is immediate: E1 already contains a topological geometric three-dimensional family naturally tied to the Lorentz object rather than chosen arbitrarily.

14 The E1 future shell and its tangent three-space

Experiment 1 defines the future unit shell

$$\mathcal{H} = \{u \in \mathcal{S} : N_{\mathcal{S}}(u) = 1, \text{ } u \text{ future}\}.$$

Its shell branch proves

$$T_u \mathcal{H} = u^{\perp_{B_{\mathcal{S}}}}, \tag{7}$$

and defines the positive tangent metric

$$g_u(v, w) = -B_{\mathcal{S}}(v, w). \tag{8}$$

The restriction is positive definite because u is timelike. Tasks 20–23 then construct the tangential projection, projected metric connection, explicit geodesics, intrinsic distance

$$d_{\mathcal{H}}(u, v) = \text{arcosh } B_{\mathcal{S}}(u, v),$$

and prove that this distance recovers the already existing shell topology, followed by metric completeness [12].

Therefore the assignment

$$u \mapsto E_u := T_u \mathcal{H} = u^{\perp_{BS}} \quad (9)$$

is a canonical candidate for an E2 metric-oriented three-dimensional fibre family, but now with topology and variation inherited from a pre-existing geometric object.

15 Established verification: this is the standard hyperboloid rest-space construction

Established Lorentzian and hyperbolic geometry gives exactly the same result. In the hyperboloid model

$$H^3 = \{u : \langle u, u \rangle_L = -1, u^0 > 0\},$$

differentiating the level-set equation gives

$$T_u H^3 = u^{\perp}.$$

The Lorentz form restricted to $u^{\perp}$ is positive definite under the corresponding signature convention, producing the Riemannian metric on hyperbolic space [9, 10, 8]. Thus the E1 shell theorem is not a novel object but an independently reconstructed standard one.

For E2 the important point is the dependency direction. The three-dimensional Euclidean fibre is no longer arbitrary input:

$$(\mathcal{S}, B_S, \mathcal{H}, u) \implies \boxed{E_u = u^{\perp}} \implies \boxed{\text{positive metric on } E_u}.$$

This is precisely the sort of deeper structural source that the arbitrary `MetricOrientedRankThreeFamily` abstraction had forgotten.

16 Orientation: an exact bridge still has to be proved

E2's fibres are not merely metric three-spaces; each carries an orientation. E1 tracks both Lorentz orientation and future-component choices, and its Hodge branch makes the orientation dependence explicit. In established geometry, an oriented time-oriented Lorentz four-space together with a future unit timelike vector u induces an orientation on $u^{\perp}$: a basis (e_1, e_2, e_3) of $u^{\perp}$ is positive when (u, e_1, e_2, e_3) has the chosen spacetime orientation, with the convention adjusted to the project's ordering/sign policy.

This yields a specific cross-experiment theorem target:

$$\text{orient}_{E2}(E_u) \stackrel{?}{=} \text{orient}_{E1}(u^{\perp} \text{ induced by } u \text{ and the Lorentz orientation}). \quad (10)$$

Equation (10) is established mathematics but has not yet been formalized across the two projects. The sign convention matters because E1 already discovered independent orientation/pseudoscalar sign choices in the Hodge–Clifford bridge.

17 Established group reconvergence: the visible E2 group

The proper orthochronous Lorentz group acts transitively on the future unit hyperboloid. Fix $u_0 \in H^3$. The stabilizer of u_0 consists exactly of the spatial rotations of $u_0^{\perp}$, hence is isomorphic to $SO(3)$. Equivalently,

$$H^3 \cong SO^+(1, 3)/SO(3). \quad (11)$$

This is standard homogeneous-space geometry [9, 10].

Experiment 2 independently reconstructed a visible group `GvisModel` and later identified it with the ordinary orientation-preserving orthogonal group of a three-dimensional Euclidean model. Therefore the first serious group bridge is not merely

$$G_{\text{vis}} \cong SO(3),$$

but the more informative dependency statement

$$\boxed{G_{\text{vis}}^{E2} \stackrel{?}{\cong} \text{Stab}_{G_L^{E1}}(u_0).} \quad (12)$$

The right-hand side is derived from the E1 Lorentz action; the left-hand side was reconstructed independently from E2's spatial transformation structure. Proving (12) would be the first genuine cross-experiment reconvergence theorem.

18 The frame bundle of the E1 shell is already the Lorentz group

The relation is stronger than stabilizer equivalence. For the standard hyperboloid model, fix a positively oriented Lorentz orthonormal basis (e_0, e_1, e_2, e_3) with $e_0 = u_0$. Then

$$\Lambda \in SO^+(1, 3) \mapsto (\Lambda e_0; \Lambda e_1, \Lambda e_2, \Lambda e_3) \quad (13)$$

identifies $SO^+(1, 3)$ with the oriented orthonormal frame bundle $\text{Fr}^+(H^3)$; this identification is used explicitly in the established hyperbolic-dynamics literature [2]. The right $SO(3)$ action changes only the spatial frame while leaving the base point Λe_0 fixed.

This is remarkably close to the global E2 object downstairs. E2 has spent Tasks 24–32 reconstructing oriented frame torsors, varying frame families, continuous transition systems and their topology. For the E1-derived shell family, established mathematics says that the entire ordinary frame total space can be presented globally as the Lorentz group itself.

The appropriate comparison target is therefore

$$\boxed{E^2 \mathcal{F}^+(E = T\mathcal{H}) \stackrel{?}{\cong} G_L^{E1}} \quad (14)$$

with the base map corresponding to $\Lambda \mapsto \Lambda u_0$ and the right visible action corresponding to the stabilizer action.

19 The E1 spin cover and the E2 internal lift

Experiment 1 also independently constructs a Clifford/spin branch and only later identifies its spin group with the standard proper Lorentz spin group. Established mathematics gives

$$\text{Spin}^+(1, 3) \cong SL(2, \mathbb{C}), \quad \text{Spin}(3) \cong SU(2),$$

and

$$H^3 \cong SL(2, \mathbb{C})/SU(2). \quad (15)$$

The projection

$$\text{Spin}^+(1, 3) \longrightarrow SO^+(1, 3)$$

has kernel $\{\pm 1\}$, and the preimage of the timelike stabilizer $SO(3)$ is $\text{Spin}(3) \cong SU(2)$ [5, 4].

Experiment 2 independently reached a group `Lift` equivalent to the unit quaternions/ $SU(2)$, a visible group equivalent to $SO(3)$, and a certified two-element central kernel. Thus established mathematics predicts the second bridge

$$\boxed{\mathcal{L}^{E2} \stackrel{?}{\cong} p^{-1}(\text{Stab}_{G_L^{E1}}(u_0)) \cong \text{Spin}(3) \cong SU(2),} \quad (16)$$

with a commuting projection to (12) and the same kernel $\{\pm 1\}$.

This relation should *not* be imported into E2 as an assumption before the bridge test. It is exactly the independent reconvergence we want to verify.

20 The established target diagram

The conventional geometry suggests the following diagram for the E1 shell specialization:

$$\begin{array}{ccc}
 \text{Spin}^+(1, 3) \cong SL(2, \mathbb{C}) & \xrightarrow{2:1} & SO^+(1, 3) \cong \text{Fr}^+(H^3) \\
 \downarrow \text{right fibre} & & \downarrow \text{right fibre} \\
 \text{Spin}(3) \cong SU(2) & \xrightarrow{2:1} & SO(3) \\
 & \searrow & \swarrow \\
 & H^3 \cong \mathcal{H} &
 \end{array} \tag{17}$$

More precisely, $\text{Spin}^+(1, 3)$ and $SO^+(1, 3)$ are total spaces over H^3 with fibres $\text{Spin}(3)$ and $SO(3)$, and the horizontal map is equivariant with respect to the two-fold cover.

The E2 construction is trying to produce exactly the same *shape* from local frame/lift data:

$$\mathcal{I} \longrightarrow \mathcal{F}^+, \quad \text{right fibre } \mathcal{L} \rightarrow G_{\text{vis}}, \quad \ker = \{\pm 1\}.$$

The present evidence therefore points to a concrete cross-experiment hypothesis:

$$\boxed{(\mathcal{I}^{E2}, {}^{E2}\mathcal{F}^+, \mathcal{L}^{E2}, G_{\text{vis}}^{E2}) \stackrel{?}{\cong} ({}^{E1}\text{Spin}^+(1, 3), G_L^{E1}, \text{Spin}(3), SO(3))} \tag{18}$$

when E2 is instantiated on the E1 shell tangent family. Equation (18) is *not yet a formal theorem*. It is the exact established-mathematics comparison target that now justifies a bridge project.

21 Why the E1 specialization may remove the apparent global obstruction

There is another established fact with major interpretive consequences. Hyperbolic three-space is diffeomorphic to $\mathbb{R}^3$ and hence contractible. Its tangent bundle is trivial, and it admits spin structures; on a simply connected contractible base the standard obstruction vanishes and the spin structure is unique up to the usual equivalence [10, 5].

Therefore, if the physically relevant E2 family is indeed the E1 shell tangent family, one should *expect* the family-level liftability problem to be positive. This does not make Tasks 28–32 wasted work. Those tasks exposed precisely which obstruction-like phenomena can occur for arbitrary presentations and arbitrary families, and they identified the topology required to state the genuine global question correctly. The E1 specialization may now collapse the broad obstruction problem because the common Lorentz source selects a much narrower geometry.

A particularly attractive intrinsic proof route would avoid characteristic classes entirely: construct the global ordinary frame object from G_L using (13), construct the upstairs object from E1's intrinsic spin cover, and compare the resulting fibre action to E2's independently reconstructed $\mathcal{L} \rightarrow G_{\text{vis}}$. If this closes, the spatial branch would be solved on the actual E1-derived geometry by explicit structure rather than by a general obstruction theorem.

22 A crucial semantic boundary: the shell is not yet spacetime

The apparent match has one important limitation. The E1 future shell $\mathcal{H}$ is the space of future unit timelike directions in a fixed Lorentz vector space. It is naturally interpreted as a velocity/observer hyperboloid or hyperbolic homogeneous space. It is not automatically the spacetime base manifold.

Thus

$$u \mapsto u^\perp$$

is a canonical spatial-rest-space family over *observer directions*. If E2's eventual intended base is spacetime itself, an additional map or field is required. In standard Lorentzian geometry a unit timelike vector field $u(x)$ on spacetime produces a spatial distribution

$$E_x = u(x)^\perp \subset T_x M,$$

and corresponds to a reduction of the Lorentz frame structure toward the spatial stabilizer. A time orientation alone does not select such a global observer field.

Proof boundary 3 (Do not confuse the first merge with the final physical base). *The E1 shell tangent family is the correct first mathematical cross-check because it is canonically generated from the common Lorentz object and exactly matches the dimensional/group data of E2. It does not by itself establish that E2's arbitrary base B should be identified with spacetime, nor does it derive a global observer field in a curved Lorentzian manifold.*

This distinction makes the shell specialization ideal as a smoke test for the merger: it is strong enough to test topology, frames and lifts, but narrow enough not to import general-relativistic structure prematurely.

23 Which E1 structures are needed now, and which are not

The current bridge should consume the smallest possible E1 interface. The following table separates immediate necessities from downstream comparison material.

E1 structure	Immediate status	Reason
Positive Euclidean $V = \mathbb{R}^3$	REQUIRED interface	Candidate identification with E2's fixed metric model; orientation convention must be checked.
Lorentz carrier $\mathcal{S}$ and bilinear form $B_{\mathcal{S}}$	REQUIRED interface	Defines the shell and rest spaces $u^\perp$.
Future shell $\mathcal{H}$	REQUIRED interface	Natural topological base for the first cross-experiment smoke test.
Shell tangent theorem $T_u \mathcal{H} = u^\perp$	REQUIRED interface	Produces the varying three-dimensional fibres without arbitrary Choice.
Positive tangent metric $-B_{\mathcal{S}}$	REQUIRED interface	Supplies the E2 fibre metric from E1 structure.
Shell topology / smooth structure	REQUIRED interface	Prevents E2 from regenerating an arbitrary total-space topology.
Lorentz group G_L and action on $\mathcal{H}$	REQUIRED comparison	Needed to test stabilizer $\leftrightarrow G_{\text{vis}}$ and frame-total $\leftrightarrow G_L$.
Orientation data	REQUIRED, convention-sensitive	Needed to instantiate E2's metric-oriented family exactly.
Jordan product	NOT needed as bridge input	Important provenance upstream of Lorentz data, but importing it into E2 would add ballast to the immediate topology/frame test.
Hodge operator	DEFERRED comparison	May later verify orientation/complex-structure reconvergences; not needed to define $T\mathcal{H}$.

E1 structure	Immediate status	Reason
Clifford algebra	FIREWALLED initially	E2's lift should remain independently reconstructed during the first bridge.
E1 spin cover	COMPARISON-ONLY initially	Use only after E2's own $\mathcal{L}$ interface is frozen; then test the commuting double-cover diagram.
Field/connection branch beyond shell Levi-Civita data	OUT OF SCOPE	Would mix dynamics/gauge questions into a topology/frame interface task.

This separation is essential. The fact that E1 already knows a spin cover is precisely why it must not be imported as construction input into the E2 lift branch. Otherwise the strongest future reconvergence would become circular.

24 E1/E2 comparison against established mathematics

The independent experiments and the standard objects can now be aligned without claiming a formal merge.

Experiment 1	Experiment 2	Established mathematics	Current bridge status
Positive Euclidean V	E2 Model	Oriented Euclidean 3-space	Expected equivalence; not cross-formalized.
$\mathcal{H}$	arbitrary base B	Hyperbolic H^3 for the shell model	Proposed E2 specialization only.
$T_u \mathcal{H} = u^\perp$	fibre E_b	Tangent/rest space of hyperboloid	Exact established match; cross theorem missing.
$-B_S _{u^\perp}$	fibre inner product	Hyperbolic tangent metric	Exact established match.
Induced spatial orientation	F.orient	Orientation of tangent 3-space	Convention-sensitive theorem needed.
$\text{Stab}_{G_L}(u)$	G_{vis}	$SO(3)$	High-priority group bridge.
G_L over $\mathcal{H}$	E2 ordinary frame total	$\text{Fr}^+(H^3) \cong SO^+(1,3)$	High-priority global bridge.
E1 $\text{Spin}^+(1,3)$	E2 global internal total	Spin frame total over H^3	Comparison-only after first bridge.
$\text{Stab}_{\text{Spin}^+(1,3)}$	E2 $\mathcal{L}$	$\text{Spin}(3) \cong SU(2)$	Strong independent reconvergence target.
Kernel $\{\pm 1\}$	E2 kernel $\{\pm 1\}$	Standard spin double cover	Exact expected match.
E1 shell topology	E2 atlas-generated topology	Tangent/vector-bundle topology	Equality must be proved, not assumed.

Experiment 1	Experiment 2	Established mathematics	Current bridge status
E1 Clifford/Hodge	E2 spin-like/lift data	Clifford-spin/Hodge framework	Later structural comparison, not immediate input.

The table shows why the connection is now mathematically compelled. E2 is not merely “similar” to E1. Its missing structural inputs sit exactly at nodes that E1 already reconstructed from the common starting object.

25 The merge problem is now an engineering theorem boundary

The mathematical cross-target is clear enough to formulate, but the projects are not yet in a form that guarantees a clean formal import.

The Task-32 artifact is a complete Lake project:

```
name = "RequestProject"

[[require]]
name = "mathlib"
git = "https://github.com/leanprover-community/mathlib4.git"
rev = "v4.28.0"

[[lean_lib]]
name = "RequestProject"
globs = ["RequestProject.+"]
```

with toolchain leanprover/lean4:v4.28.0.

The public E1 branch, at commit 03d16306a214512cfa6e675f348ceb5a3e89494a, contains the cumulative papers and a compressed Aristotle archive, but not an unpacked Lake project at the repository root [11]. The final E1 reports state that the formal development used Lean 4.28.0 and modules under `RequestProject`, which is encouraging for toolchain compatibility, but the package configuration inside the published compressed archive has not been independently extracted in the present environment.

Engineering finding 1 (Current importability status). Lake itself supports dependencies from local paths and Git repositories, including a `subDir` for a package stored below a repository root [7, 6]. Therefore there is no fundamental Lean/Lake obstacle to a two-package bridge. The unresolved issue is practical: whether the Aristotle project submission environment can access or resolve the second project in the required form, and whether the existing E1 archive can be exposed as a proper Lake dependency without refactoring. This has not yet been tested.

26 Why refactoring may be necessary even if imports work

A technically successful import of the complete E1 and E2 projects would still be scientifically undesirable. Each project contains a long historical proof tree. Importing both wholesale would make it easy to consume a downstream result accidentally and destroy the experimental independence we have protected for more than thirty tasks.

There is also a package-engineering risk. E2’s package is named `RequestProject`; E1’s reports use the same module root and may also use the default Aristotle package naming convention. Lake requires package names to be unique in a dependency graph. The exact E1 package configuration

must therefore be inspected before a merge. Even if package names do not collide, identical generic module roots can make human auditing harder.

The correct response is not to rewrite the mathematics. It is to create narrow export layers.

27 Recommended refactoring architecture

The frozen original projects should remain immutable. Additive interface packages can then expose only certified endpoints.

Experiment-1 interface package

A minimal package, for example `Experiment1Interface`, should export only:

1. the positive Euclidean model V and its metric/orientation interface;
2. $\mathcal{S}$, $B_{\mathcal{S}}$, $N_{\mathcal{S}}$ and the future-shell carrier;
3. the theorem $T_u \mathcal{H} = u^\perp$ or the exact formal substitute used in E1;
4. positivity of $-B_{\mathcal{S}}$ on the shell tangent space;
5. the shell topology/smooth interface;
6. the Lorentz group action on the shell.

It should *not* initially export the E1 Clifford algebra, spin cover, $SL(2, \mathbb{C})$ identification, Hodge sectors, or field/gauge branch.

Experiment-2 interface package

A minimal `Experiment2SpatialInterface` should export only:

1. the E2 metric-oriented three-space model and visible group;
2. the independently reconstructed `Lift` and projection to `Gvis`;
3. the frame torsor/frame-family interface;
4. the global gluing interface from Tasks 29–31;
5. the Task-32 ordinary atlas/refinement relations needed for topology comparison.

Historical exploratory modules and comparison labels should remain behind the interface.

Bridge package

A third package, for example `LorentzSpatialBridge`, should require both interfaces and contain no new primitive physics. Its first job is compilation, not theorem discovery.

28 Three viable import strategies

Strategy	Advantages	Risks / required test
Lake local-path dependencies	Clean development setup; frozen projects remain separate; easiest for local smoke tests.	Aristotle cloud submission may not receive sibling paths; must test packaging/upload behavior.

Strategy	Advantages	Risks / required test
Git dependencies with <code>subDir</code>	Reproducible by commit; natural long-term architecture.	E1 repository must first contain an unpacked Lake package/subdirectory; current public branch does not.
Single vendored bridge project	Most likely to work in an upload-only environment; one toolchain and one manifest.	Highest contamination risk; package/module namespaces must be separated and source provenance must remain exact.

The first formal experiment should choose the least invasive strategy that Aristotle actually accepts. No mathematical bridge theorem should be attempted until the packaging test itself is green.

29 Minimal import smoke test

The smoke test should be intentionally trivial:

1. pin Lean 4.28.0 and the same Mathlib revision used by E2;
2. expose one E1 theorem or definition from the shell interface;
3. expose one E2 theorem or definition from the spatial interface;
4. create a new bridge module importing exactly those two modules;
5. prove only two identity/well-typedness lemmas, one consuming each side independently;
6. run `#printaxioms`;
7. run the focused bridge build and then the complete bridge workspace build.

A successful smoke test proves only that the projects can coexist technically. It must not yet identify any E1 and E2 objects.

30 The first mathematical bridge after the smoke test

Once imports are known to work, the first cross theorem should remain below the spin level. Fix the E1 future shell and define

$$E_u := T_u \mathcal{H} \cong u^\perp.$$

The bridge should prove that this data instantiates E2's metric-oriented rank-three family while preserving the E1 topology rather than allowing E2 to generate a new arbitrary one.

The hard targets are:

1. **carrier:** construct the E2 fibre family from the E1 shell tangent/rest-space family;
2. **rank and metric:** prove rank three and identify the E2 inner product with $-B_S$;
3. **orientation:** prove the exact induced-orientation convention;
4. **topology:** define the total tangent/rest-space carrier with the E1-induced topology and prove any accepted E2 atlas is admissible for that fixed topology;
5. **topology invariance:** prove ordinary gauge/refinement equivalence preserves the fixed topology rather than creating a new one.

Only after these close should the visible-group bridge (12) be attempted.

31 The second bridge: visible stabilizer and frame total

The next stage should compare E2's visible spatial group with the E1 Lorentz stabilizer. It should not start from the standard isomorphism $SO(3)$ but derive the map from actions:

1. an E1 Lorentz transformation fixing u_0 restricts to an orientation-preserving isometry of $u_0^\perp$;
2. this produces a homomorphism into E2's visible fibre group;
3. injectivity follows because a Lorentz transformation fixing u_0 and acting trivially on $u_0^\perp$ is the identity;
4. surjectivity should be constructed by extending a spatial orientation-preserving isometry by the identity on the timelike line.

This would prove the stabilizer bridge internally and recover the standard $SO(3)$ result as a late comparison.

Then one can globalize the map to compare G_L with the E2 oriented frame total space over $\mathcal{H}$ using the Lorentz action on a reference orthonormal frame.

32 Only then compare the two spin/lift branches

After the spatial geometry and visible group have been identified, freeze the E2 **Lift** interface and separately expose the E1 spin-cover comparison interface. The final cross theorem should ask whether there exists an isomorphism making

$$\begin{array}{ccc} \mathcal{L}^{E2} & \longrightarrow & \text{Spin}(3)^{E1} \\ p \downarrow & & \downarrow \\ G_{\text{vis}}^{E2} & \longrightarrow & \text{Stab}_{G_L^{E1}}(u_0) \end{array}$$

commute, with kernel correspondence $\{\pm 1\} \leftrightarrow \{\pm 1\}$.

If that local stabilizer diagram closes, the global total-space comparison (18) becomes a legitimate next theorem. The point of the ordering is crucial: E1's known spin cover is used only as a *comparison target* after E2's independently derived lift is frozen.

33 What happens to Clifford, Hodge and Jordan at the junction

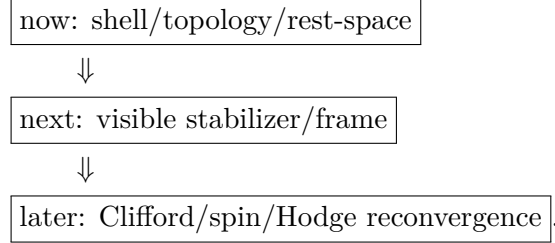
The earlier E1 excavation found Jordan, Hodge and Clifford structures. Their roles at the present junction are different.

Jordan. The spin-factor/Jordan product is upstream provenance for the E1 Lorentz norm and future cone. It explains how the Lorentz shell arose in E1, but E2 does not need the Jordan product to instantiate the shell tangent family. It should therefore remain behind the E1 interface for the first bridge.

Clifford. Clifford structure is highly relevant to the eventual comparison because E1's spin cover is constructed through that branch and E2's **Lift** has the expected $Spin(3)/SU(2)$ form. But precisely for that reason Clifford must be firewalled from the initial E2 construction interface; otherwise the independent discovery is lost.

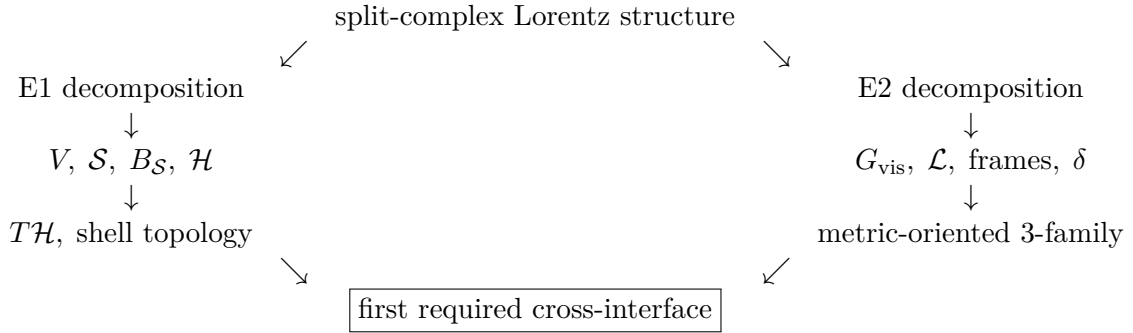
Hodge. Hodge structure may become the cleanest bridge for orientation and for the complex structure on the Lorentz bivector/Lie algebra, but it is not required to solve the immediate total-space-topology problem. Its first legitimate role is therefore a later consistency test: does the orientation chosen in the rest-space bridge reproduce the Hodge/Clifford convention already tracked in E1?

This yields a useful hierarchy:



34 Updated dependency DAG: discovery versus reconstruction

The top-to-bottom discovery graph now contains two independently excavated shafts from the same Lorentz starting object:



This graph is a *discovery* DAG. The eventual foundational reconstruction direction must be audited separately. If the cross-interface closes, it may reveal that some E2 variables were already derivable from deeper E1 variables, but this cannot be assumed merely because the paths meet.

The bottom-up reconstruction candidate would then begin with the common deeper data and ask which subsets reconstruct the shell, spatial frame group, internal lift and ultimately the original Lorentz structure. That is a later task.

35 Comparison with established spin-bundle mathematics

Standard spin geometry says that an oriented Riemannian rank- n vector bundle has an oriented orthonormal frame bundle with structure group $SO(n)$, and a spin structure is an equivariant lift of that principal bundle through the two-fold cover $Spin(n) \rightarrow SO(n)$ [5, 4]. From this perspective, the E2 sequence is recognizable:

$$\begin{aligned}
 \text{metric-oriented fibres} &\rightarrow \text{oriented frames} \rightarrow SO(3)\text{-valued transitions} \\
 &\rightarrow SU(2)\text{-valued local lifts} \rightarrow \{\pm 1\}\text{-defect} \\
 &\rightarrow \text{global lifted total space.}
 \end{aligned}$$

The striking point is not that the endpoint resembles known spin geometry; that is expected. The useful information is where E2's dependencies were too weak or too strong:

- too strong: demanding whole-overlap lifts for one fixed arbitrary atlas;
- too weak: replacing a topological bundle-like family by a pointwise collection of fibres and then quantifying over all newly generated atlas topologies.

Established theory therefore supports the source-review diagnosis: the invariant should attach to a fixed topological vector/frame bundle, not to one arbitrary cocycle presentation and not to a topology-free family of fibres.

36 Formal and mathematical decision table after Part XXXII

Question	Status	Scope
Ordinary chart-comparison maps reconstructed?	PROVED	Task 32.
Exact ordinary transition gauge law?	PROVED	$g'_{ij} = h_j g_{ij} h_i^{-1}$.
Ordinary cover refinement formalized?	PROVED	Including identity/composition.
Common ordinary refinement?	PROVED	Intersection construction.
Generated ordinary atlas equivalence?	PROVED	Finite zig-zag relation.
Task-31 good/bad presentations on same bare family?	PROVED	Mandatory control.
Good/bad chart gauge $h_0 = 1, h_1 = b$?	PROVED	Exact source theorem.
Arbitrary ordinary chart gauges preserve direct liftability?	NO	Counterexample survives.
Internally liftable ordinary gauges preserve direct liftability?	PROVED	Strongest correct transfer.
Direct liftability passes to refinement?	PROVED	Restriction theorem.
Refinement converse?	REFUTED	Bad atlas has liftable refinement.
Task-32 <code>FamilyLiftable</code> defined?	PROVED	Existential over any atlas of bare family.
<code>FamilyLiftable iff FamilyKernelDefectNeutral</code> ?	PROVED	Exact at current definition.
Family-level defect quotient across all covers?	NOT PACKAGED	Boolean existential only.
Bare family contains fixed total-space topology?	NO	Definitionally absent.
Atlas equivalence proved to preserve total-space topology?	NOT PROVED	Principal missing theorem.
Current family-neutrality theorem uses atlas-equivalence witness?	NO	Source-level review.
Universal family liftability under current definition?	FORMALLY OPEN	Source-level route strongly suggests YES.
Genuine non-liftable topological frame geometry?	NOT ESTABLISHED	No witness.
E1 shell topology reconstructed independently?	PROVED IN E1	Shell distance and Alexandrov reconvergences.
E1 shell tangent fibres are 3D positive rest spaces?	PROVED IN E1	$T_u \mathcal{H} = u^\perp$, metric $-B_S$.
Established $H^3 \cong SO^+(1,3)/SO(3)$?	STANDARD	External comparison only.
Established $H^3 \cong SL(2, C)/SU(2)$?	STANDARD	External comparison only.

Question	Status	Scope
Established $\text{Fr}^+(H^3) \cong \text{SO}^+(1, 3)$?	STANDARD	External comparison only.
E1/E2 rest-space family bridge?	NOT YET FORMALIZED	First required cross theorem.
E1 Lorentz stabilizer $\leftrightarrow$ E2 G_{vis} ?	NOT YET FORMALIZED	High-priority bridge.
E1 spin stabilizer $\leftrightarrow$ E2 $\mathcal{L}$?	NOT YET FORMALIZED	Comparison-only after freeze.
Lake supports local/Git package dependencies?	DOCUMENTED	Lean build-system level.
Aristotle accepts required cross-project dependency setup?	UNTESTED	Engineering frontier.
Current E1 GitHub branch directly importable as Lake package?	NO AT REPO ROOT	Lean sources remain inside archive.
Refactoring/interface extraction needed?	LIKELY	Must be decided by smoke test.
Spatial spin closure?	NO	Junction reached, not closure.
Experiment-1/Experiment-2 full marriage?	NO	Only interface investigation begun.
Physical spin/dynamics/GR?	NO	Out of scope.

37 Conservation of Difficulty at the junction

The current sequence illustrates the methodological principle more sharply than any earlier task. Task 31 found a non-neutral defect. Task 32 correctly showed that this defect can disappear after ordinary refinement and correctly separated fixed-presentation liftability from existential family liftability. The apparent difficulty then seemed to disappear entirely at the family level.

Source review shows where it went:

the topology itself was no longer fixed.

The obstruction was not solved; the variable on which it could depend had been removed from the object.

Experiment 1 now enters not as an external rescue but as an independently excavated branch from the same starting structure that already contains a candidate for precisely this missing variable. If the E1 shell/rest-space topology reconstructs the E2 topological frame family without arbitrary choices, the difficulty has not been imported from outside. It has been located in another branch of the same decomposed Lorentz equation system.

This is exactly the situation the dual-experiment design was meant to detect.

38 What Part XXXIII should and should not do

Part XXXIII should begin with an engineering result, not a mathematical grand theorem.

Stage 0: importability smoke test

Determine whether Aristotle can compile a project that depends on two separately frozen Lean packages. Test local path, Git/subdirectory, or a vendored bridge package as available. Record every failed import/build exactly.

Stage 1: refactor only if necessary

If direct cross-project dependency is unavailable, export minimal E1 and E2 interface packages with distinct package names and pinned Lean/Mathlib versions. Preserve original theorem bodies and provenance; do not rewrite proof content merely to merge trees.

Stage 2: topology-preserving E1 rest-space instance

Construct the family $u \mapsto T_u\mathcal{H} = u^\perp$ as the first E2-compatible metric-oriented family while retaining the E1-derived total-space topology. Prove rank, metric and orientation compatibility.

Stage 3: ordinary atlas topology theorem

Show that any E2 atlas admitted for this fixed family is admissible for the pre-existing E1 topology and that ordinary gauge/refinement steps preserve that topology. This repairs the exact source-review flaw of Task 32.

Stage 4: visible-group bridge

Prove the stabilizer/visible-group equivalence and then the frame-total/Lorentz-group equivalence.

Stopping rule

Stop before importing E1's spin cover into the E2 construction. Freeze the visible spatial bridge first. The E1 spin cover should enter only in the subsequent comparison task, where it can test the independently reconstructed E2 lift.

39 Formal prompt boundary for the next task

The next Aristotle prompt should therefore be narrower than a full “merge E1 and E2” request. It should contain two separate success conditions:

1. **engineering success:** the bridge project imports the minimal frozen interfaces and builds green;
2. **mathematical success:** the E1 shell tangent/rest-space family is instantiated as a fixed-topology E2 metric-oriented family without consuming the E1 Clifford/spin branch.

If the first condition fails, the task should stop with a refactoring report rather than attempt mathematics in copied prose. If the second condition fails, the exact type/topology mismatch should become the next research question.

40 Conclusion

Task XXXII looked like the step that would finally quotient ordinary presentation dependence out of the E2 kernel-defect problem. It did something more useful. It formally reconstructed ordinary chart changes and cover refinements, proved the good/bad atlas comparison on one bare family, identified the exact condition under which ordinary gauges preserve direct liftability, and showed that refinement can repair a non-liftable fixed atlas. These results are genuine and remain part of the final dependency graph.

The attempted family-level closure then took a productive wrong turn. `FamilyLiftable` and `FamilyKernelDefectNeutral` are existential predicates on a topology-free collection of metric-oriented fibres. Their equivalence is formally correct, but the abstraction has erased the total-space topology that Task XXVII had introduced precisely because Task XXVI could not derive it from pointwise fibre data. The inherited Choice trivialization theorem strongly suggests that universal

family liftability is actually trivial at this level: choose a global algebraic chart and let that chart generate the topology. A future hardening run should make this source-level diagnosis a Lean theorem.

That diagnosis changes the research direction. Experiment 2 should not continue indefinitely into arbitrary vector-bundle obstruction theory before checking whether its missing topology and spatial family are already supplied by the independently decomposed Experiment 1. Experiment 1 contains a future unit shell with intrinsic topology, tangent spaces $T_u\mathcal{H} = u^\perp$, positive spatial metric $-B_S$, Lorentz action, Clifford/Hodge branches and an independently constructed spin cover. Established mathematics shows that these objects sit in the exact diagram

$$H^3 \cong SO^+(1, 3)/SO(3) \cong SL(2, \mathbb{C})/SU(2), \quad \text{Fr}^+(H^3) \cong SO^+(1, 3),$$

with the standard spin double cover and kernel $\{\pm 1\}$. Those are precisely the structures E2 has been reconstructing from another direction.

The correct next move is therefore neither to declare victory nor to force a known spin obstruction theorem into the project. It is to test whether the two independent Lean developments can coexist as imports, refactor them behind narrow interfaces if required, and prove the first cross-experiment theorem below the spin level: the E1 shell/rest-space geometry supplies a fixed-topology metric-oriented three-family of the kind E2 requires. Only after that bridge is green should the visible stabilizers be compared, and only after the visible bridge is frozen should E1's spin cover be allowed to meet E2's independently reconstructed lift.

The project has not reached the basement. It has reached the first structural junction at which continuing down one shaft alone is no longer obviously meaningful. Patience here is not delay; it is part of the proof boundary.

A Task-XXXII principal theorem ledger

For auditability, the following Task-32 endpoints are central to the paper's interpretation:

- `chartChange_transition_law;`
- `OrdinaryCoverRefinement.transitionFun_eq;`
- `ordinaryAtlasEquivalent_equivalence;`
- `GoodBad.same_underlying_bare_family;`
- `GoodBad.gauge_zero, GoodBad.gauge_one;`
- `GoodBad.trivialAtlas_directAtlasLiftable;`
- `GoodBad.badAtlas_not_directAtlasLiftable;`
- `GoodBad.fixedPresentation_neutrality_not_familyInvariant;`
- `directAtlasLiftable_refine;`
- `GoodBad.directLiftability_refinement_converse_refuted;`
- `internalLiftable_gaugedSystem;`
- `GoodBad.goodBadGauge_not_internallyLiftable;`
- `tripleDefect_conjugation;`
- `familyLiftable_iff_familyKernelDefectNeutral;`
- `presentationLevel_nonNeutral_with_familyLevel_neutral;`

- `familyLiftable_constructs_globalTwofoldInternalFrameLift`;
- `task32_verdictC_content`.

The inherited source-level facts essential to the review are:

- Task XXVI `MetricOrientedRankThreeFamily` contains no total-space topology;
- Task XXVI `exists_localFibreTriv` gives algebraic trivializations over every subset by Choice;
- Task XXVII `MetricOrientedFibreAtlas.top` generates the total fibre topology from an atlas;
- Task XXVII `eq_top_of_isAdmissible` proves uniqueness of an admissible topology relative to that atlas.

B Engineering provenance and next-build obligations

The next formal stage should preserve the usual engineering provenance discipline:

1. record the exact E1 archive/package configuration before modifying anything;
2. record the exact E2 package configuration already frozen at Task 32;
3. record package names, Lean toolchains, Mathlib revisions and module roots;
4. test the smallest possible cross-package import;
5. preserve every import-resolution, package-name, toolchain, dependency-manifest and namespace failure;
6. if refactoring is required, preserve the original source hashes and document every changed path/package declaration separately from theorem content;
7. do not write final bridge documentation until the focused and whole bridge workspace builds are green;
8. run `#printaxioms` on every new cross-experiment endpoint;
9. maintain a strict import firewall so that the initial bridge cannot consume E1’s Clifford/spin comparison branch.

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